

## Contextual bandit against the rule: complete results

**Protocol.** One pull is one real request on the phone: Llama-3.2-1B on the Adreno 840, 9737-token prompt, 1024 output tokens,  $K=1024$ ,  $\mu\text{KV}$  frozen config. Cooling gate ( $\text{DDR} \leq 35^\circ\text{C}$ ,  $\text{battery} \leq 33^\circ\text{C}$ ) before every pull, CPU caps pinned, GPU cap restored after each pull. Energy is metered from the USB rail plus the coulomb counter. Epsilon-greedy,  $\epsilon = 0.25$ , warm start of one pull per (context, arm). 24 pulls on 2026-09-03, 11:08 to 13:37.

**Reward.**  $r = w_q - w_t T/T_{\text{ref}} - w_e E/E_{\text{ref}}$  with  $E_{\text{ref}} = 782.0\text{ J}$  and  $T_{\text{ref}} = 142.4\text{ s}$ , the cost table’s prediction for this request at 1200 MHz. The weights are the scheduler’s own tier weights, so the bandit optimizes exactly the objective the rule is built to serve.

**Table R1.** Contexts and their weights. Identical to the scheduler’s lever anchors.

| Context | battery       | $w_q$ | $w_t$ | $w_e$ | time slack |
|---------|---------------|-------|-------|-------|------------|
| healthy | mains or >50% | 0.55  | 0.40  | 0.05  | 5%         |
| mid     | 21 to 50%     | 0.35  | 0.20  | 0.45  | 22%        |
| low     | $\leq 20\%$   | 0.20  | 0.10  | 0.70  | 40%        |

**Table R2.** Every context and arm the bandit pulled. Energy and time are measured means. The bandit’s preferred arm is marked. All 24 pulls are represented.

| Context | Arm (MHz) | pulls | Energy (J) | Time (s) | Mean reward |                                         |
|---------|-----------|-------|------------|----------|-------------|-----------------------------------------|
| healthy | 1200      | 6     | 804.6      | 139.1    | +0.1079     | bandit’s choice                         |
| healthy | 902       | 1     | 599.4      | 180.6    | +0.0043     |                                         |
| healthy | 726       | 1     | 559.0      | 218.4    | −0.0991     |                                         |
| mid     | 1200      | 1     | 799.5      | 139.1    | −0.3054     | bandit’s choice                         |
| mid     | 902       | 6     | 579.6      | 180.5    | −0.2370     |                                         |
| mid     | 726       | 1     | 579.5      | 218.5    | −0.2903     |                                         |
| low     | 1200      | 1     | 790.2      | 139.2    | −0.6050     | bandit’s choice<br>within 0.0015 of 902 |
| low     | 902       | 4     | 591.4      | 180.7    | −0.4562     |                                         |
| low     | 726       | 3     | 563.4      | 218.4    | −0.4577     |                                         |

**Table R3.** What learning bought. Utility is the tier’s own objective evaluated on the cost table, so rule and bandit are scored on the same scale. The rule may pick split-clock plans the bandit’s arm set does not contain.

| Context | Rule’s plan              | Utility | Bandit’s arm | Utility | Gain    | Verdict     |
|---------|--------------------------|---------|--------------|---------|---------|-------------|
| healthy | prefill 1200, decode 902 | +0.0989 | 1200         | +0.0999 | +0.0010 | tie         |
| mid     | prefill 1200, decode 726 | −0.2845 | 902          | −0.2440 | +0.0405 | bandit wins |
| low     | 902                      | −0.4551 | 902          | −0.4551 | 0       | identical   |

**Table R4.** The one disagreement, in physical units. The rule refuses the bandit’s arm because it misses the mid tier’s time budget of 173.1s, not because it ranks it lower.

| Mid tier, 9737 prompt + 1024 output | Energy (J)    | Time (s)     | Utility        |
|-------------------------------------|---------------|--------------|----------------|
| Rule: prefill 1200, decode 726      | 750.0         | 144.5        | −0.2845        |
| Bandit: flat 902                    | 589.5         | 181.4        | −0.2440        |
| Difference                          | −160.5 (−21%) | +36.9 (+26%) | +0.0405 (+14%) |

**Table R5.** What learning cost. Regret is measured against an oracle that always plays the bandit’s own final best arm. Almost all of it is paid in the mandatory warm start.

|                                                                                        | Pulls | Energy (kJ) | Regret |
|----------------------------------------------------------------------------------------|-------|-------------|--------|
| Warm start, one pull per context and arm                                               | 9     | 5.9         | 0.542  |
| Free choices                                                                           | 15    | 9.8         | 0.043  |
| Total                                                                                  | 24    | 15.7        | 0.585  |
| Wall-clock cost including the cooling gates: 149 minutes                               |       |             |        |
| The rule pays none of this. Its table is seeded from measurements taken once, offline. |       |             |        |